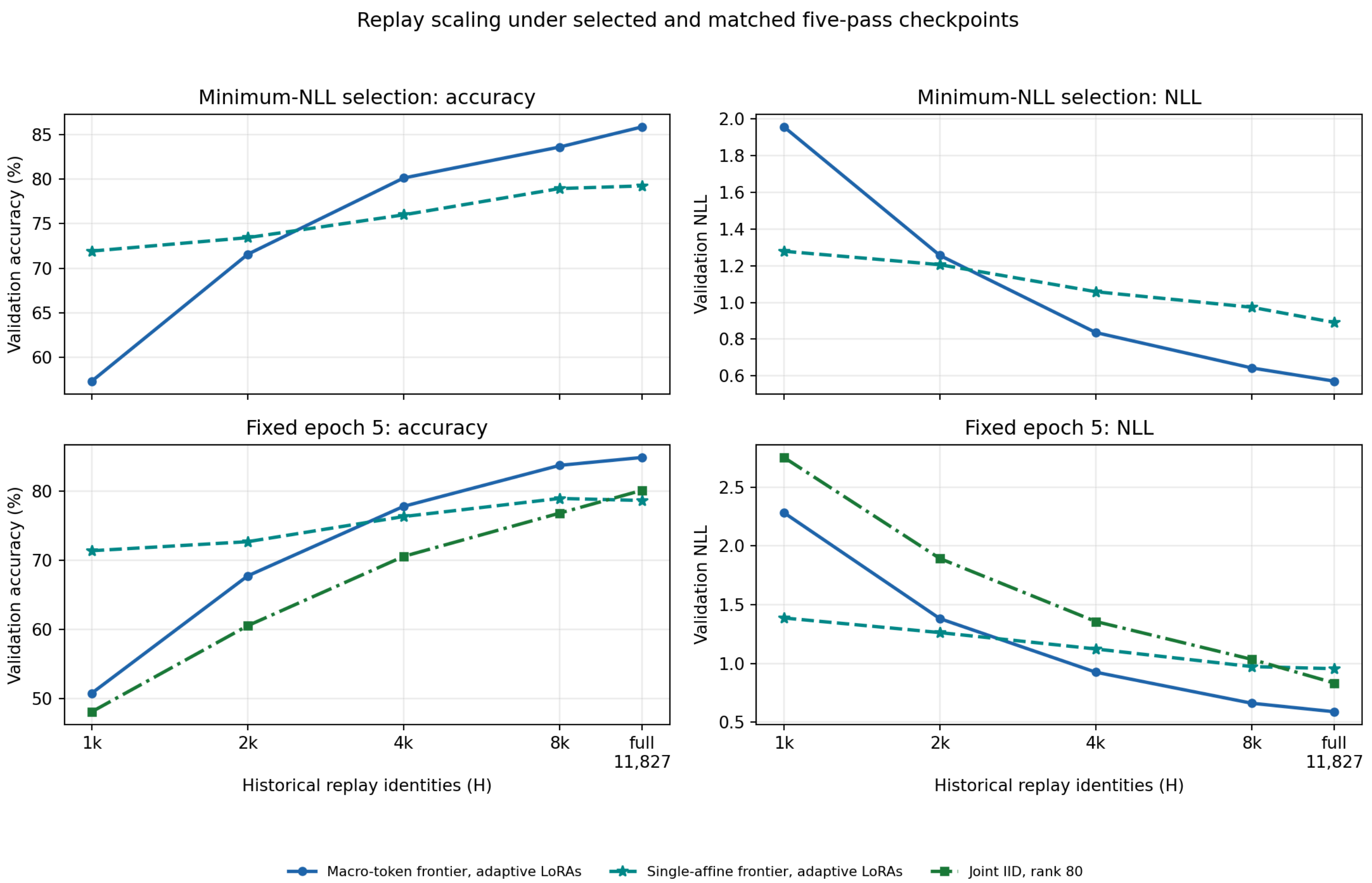
ImageNet-R stage-31 frontier-LoRA adaptation

Can jointly adapting five fragmented node representations close the same-split joint-IID accuracy and NLL gap?

Selected macro-token result. The minimum-NLL macro-token condition is Macro-token frontier, adaptive LoRAs (H=11,827; full fit): **85.864% accuracy / 0.5710 NLL**. That is 8.003 percentage points higher and 0.3715 NLL lower than the five-epoch rank-16 joint-IID reference. Macro-token frontier, adaptive LoRAs (H=4,096) is the smallest H to reach both rank-16 joint-IID metrics, first doing so at epoch 6.

Matched-H replay crossover. Under both checkpoint views, the single-affine frontier wins at H=1,024 and H=2,048; the macro-token frontier wins from H=4,096 onward. At fixed epoch five, the affine accuracy lead falls from 20.630 to 4.952 points, then the macro lead grows from 1.476 points at H=4,096 to 4.788 at H=8,192. Rank 80 trails both at every truncated H; at full history it passes the affine head but remains behind the macro head.



The upper row compares the 50-epoch minimum-NLL selections for the macro-token and single-affine frontiers. The lower row compares macro-token, single-affine, and rank-80 joint IID at fixed epoch five, after five passes over the identical fit identities at each H.

Replay scaling: selected frontier checkpoints

H	Macro-token frontier, adaptive LoRAs			Single-affine frontier, adaptive LoRAs		
	Selected epoch	Accuracy	NLL	Selected epoch	Accuracy	NLL
1,024	31	57.330%	1.9572	2	71.925%	1.2792
2,048	25	71.564%	1.2565	3	73.434%	1.2058
4,096	12	80.125%	0.8360	4	75.992%	1.0586
8,192	8	83.601%	0.6430	5	78.944%	0.9737
11,827; full fit	13	85.864%	0.5710	3	79.239%	0.8907

Macro-token and single-affine cells share the 50-epoch AdamW schedule and minimum-validation-NLL rule. Rank 80 is excluded because its frozen protocol has only five epochs and a fixed primary endpoint.

Replay scaling: fixed epoch five

H	Fit identities	Macro-token frontier, adaptive LoRAs		Single-affine frontier, adaptive LoRAs		Joint IID, rank 80	
		Accuracy	NLL	Accuracy	NLL	Accuracy	NLL
1,024	1,391	50.738%	2.2827	71.368%	1.3873	48.049%	2.7526
2,048	2,415	67.727%	1.3817	72.680%	1.2629	60.512%	1.8930
4,096	4,463	77.796%	0.9260	76.320%	1.1238	70.548%	1.3572
8,192	8,559	83.732%	0.6621	78.944%	0.9737	76.812%	1.0353
11,827; full fit	12,194	84.880%	0.5897	78.649%	0.9560	80.125%	0.8339

Within each H row, all three models see the same current-plus-history identity set for five complete passes. Their optimizer families and compute differ by design.

Full-fit architecture and capacity comparison

Condition	LoRA layout	LoRA parameters	Other trainable parameters	Total active parameters	Epoch-5 accuracy	Epoch-5 NLL
Joint IID, rank 16 (full fit; 5 epochs)	1 x 16	1,327,104	95,356	1,422,460	77.862%	0.9425
Joint IID, rank 80 (H=11,827; full fit; 5 epochs)	1 x 80	6,635,520	95,356	6,730,876	80.125%	0.8339
Joint IID, rank 224 (full fit; total-active match; 5 epochs)	1 x 224	18,579,456	95,356	18,674,812	81.109%	0.7772
Single-affine frontier, adaptive LoRAs (H=11,827; full fit) (epoch 5)	5 x 16	6,635,520	768,200	7,403,720	78.649%	0.9560
Macro-token frontier, adaptive LoRAs (H=11,827; full fit) (epoch 5)	5 x 16	6,635,520	12,055,496	18,691,016	84.880%	0.5897

Every row uses exactly five complete passes over the 12,194-image full-fit population. "Other" means the trainable joint classifier, affine integrator, or macro integrator; frozen parameters are omitted. Rank 80 matches aggregate node-LoRA capacity. Rank 224 is the closest total-active-parameter match.

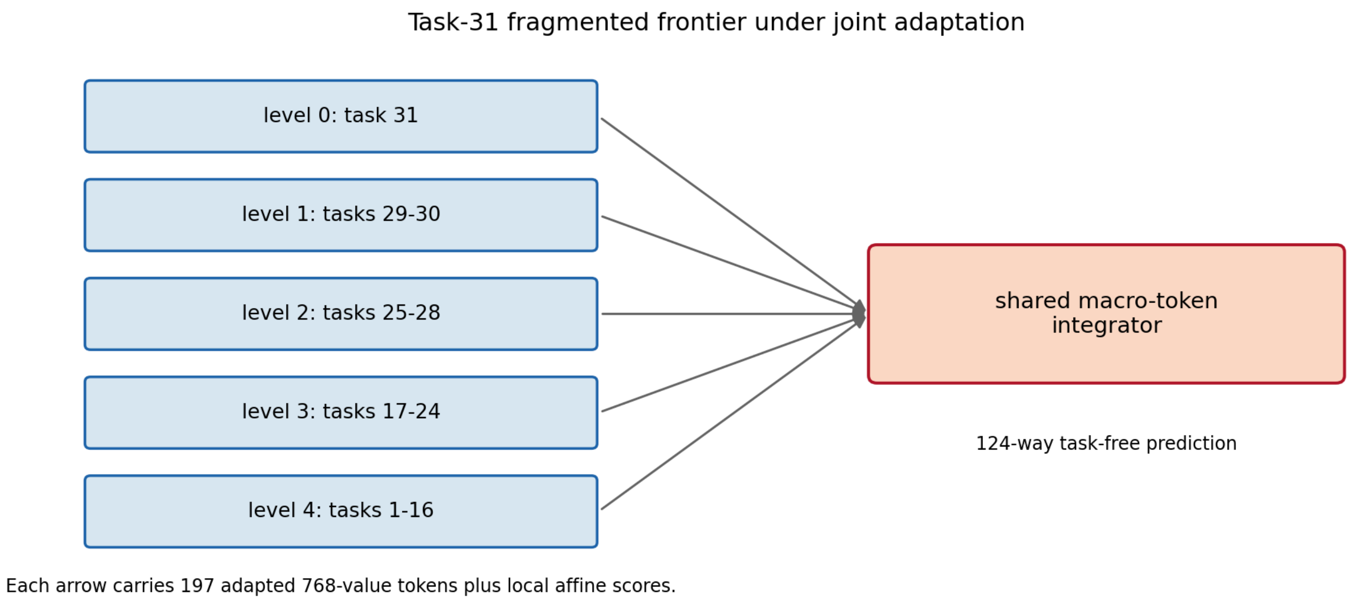
Joint-IID capacity trajectories

Epoch	Rank-80 accuracy	Rank-80 NLL	Rank-224 accuracy	Rank-224 NLL
1	75.992%	0.9977	79.272%	0.8326
2	78.485%	0.8350	81.109%	0.7311
3	80.026%	0.7954	81.699%	0.7199
4	80.354%	0.8340	82.060%	0.7727
5	80.125%	0.8339	81.109%	0.7772

Epoch five is the predeclared comparison endpoint. Minimum-NLL epochs are retained only as convergence diagnostics.

Architecture and experimental boundary

The task-31 frontier has five nodes at levels 0-4, covering task intervals 31, 29-30, 25-28, 17-24, and 1-16. Each node supplies its own final 197 x 768 LoRA-adapted token sequence and immutable local affine scores to the macro transformer. The affine frontier instead concatenates only the five 768-value pre-classifier outputs and maps the resulting 3,840 values directly to 200 logits. Neither frontier head receives task IDs or labels.



This diagram shows the macro condition: the base ViT and node classifiers are frozen, while five rank-16 LoRAs and the shared 12.06M-parameter macro head can move. The single-layer condition replaces that head with one affine map.

Every cell includes all 367 current-task images. H=1,024, 2,048, 4,096, 8,192, and 11,827 are nested prefixes of one deterministic uniform draw from the historical tasks without replacement. Maximum H therefore gives exactly all 12,194 fit identities. Frontier cells start independently from identical sealed node tensors; rank-80 cells start independently from the same seed and zero-effect LoRA initialization. The validation partition has 3,049 clean identities and never contributes gradients. No test image is opened.

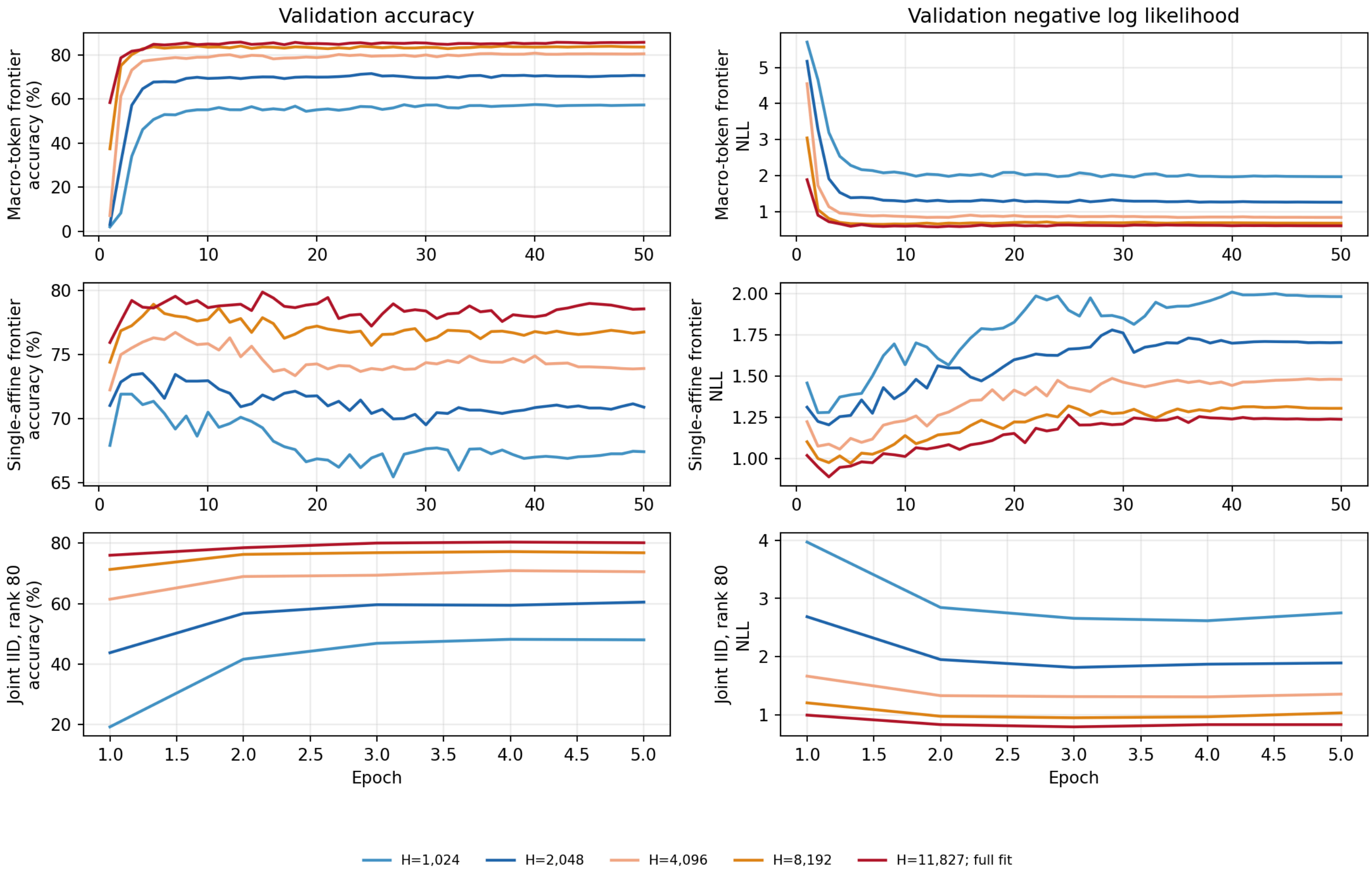
Compute boundary. Both adaptive frontier heads evaluate the same five node-specific ViTs for every image. The linear condition then runs one 768,200-parameter affine map; the macro condition runs its 12,055,496-parameter transformer. Joint IID evaluates one ViT with one shared LoRA and classifier. Their split and full-fit data exposure match, but deployment compute remains different.

Why the frozen online control exists

The previous frozen macro was trained from cached center-crop tokens. This run loads images online with deterministic random training augmentation. The purple full-fit control follows that new path while freezing the LoRAs, so its difference from the cached gray reference measures the pipeline/augmentation change rather than representation adaptation.

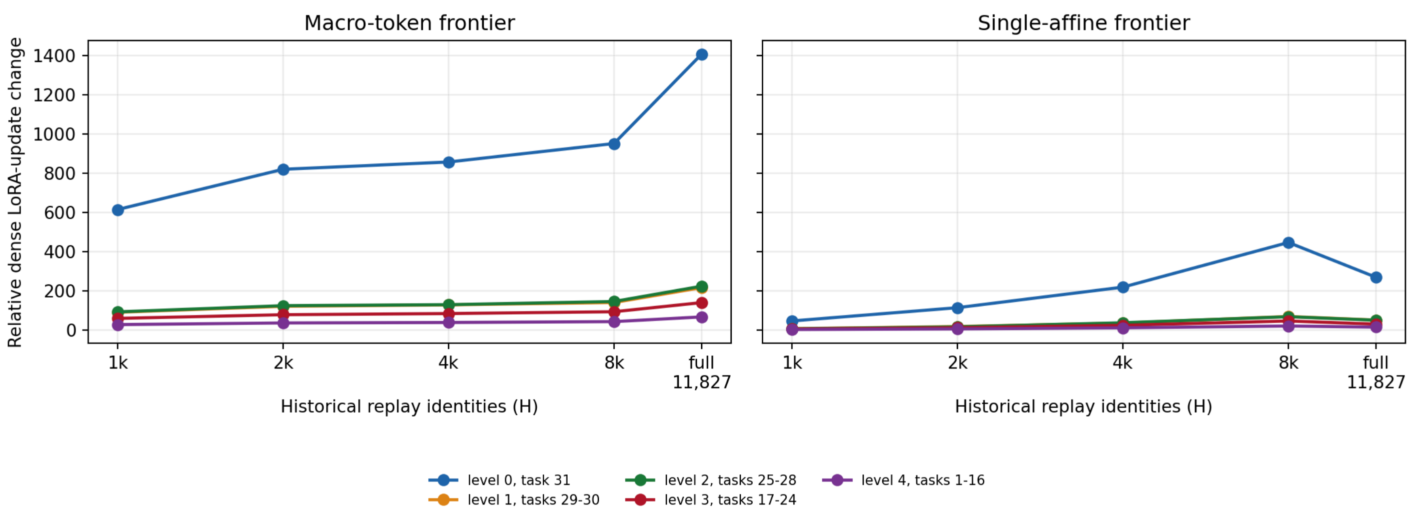
Optimization behavior

Per-architecture learning curves across replay populations



Rows separate the macro-token frontier, single-affine frontier, and rank-80 joint-IID model. Colors mean the same H in every panel; frontier histories contain 50 epochs and rank-80 histories contain the five frozen-protocol epochs.

How far each sealed frontier adapter moved



Scale-aware movement of each dense node-LoRA update from its sealed source value. Left is the macro-token frontier and right is the single-affine frontier under the same five H values.

Interpretation

Allowing the frontier LoRAs to move changes the result by 10.331 percentage points and 0.4257 NLL relative to the online frozen-LoRA control at their minimum-NLL checkpoints. H=4,096 is the smallest tested historical population to cross both rank-16 joint-IID values. At exactly 5 full-fit passes (60,970 image presentations), adaptive full history is 84.880% / 0.5897, frozen full history is 72.155% / 1.1788, and rank-16 joint IID is 77.862% / 0.9425. This isolates feature adaptation from training exposure, but not the frontier's extra model and deployment compute.

The replay-scaling tables make two different questions explicit. Minimum-NLL selection asks what each 50-epoch frontier fit can achieve after checkpoint selection. Fixed epoch five asks how the two frontiers and rank-80 joint IID compare after the same number of passes over exactly the same identities. The latter still does not equalize optimizer, initialization, parameter count beyond the LoRA-rank match, or inference compute.

What the single-layer condition establishes

The single-layer pre-classifier integrator reaches **79.239% accuracy / 0.8907 NLL** at its minimum-NLL checkpoint, epoch 3. Relative to the macro-token full-history checkpoint selected by the same rule, this changes accuracy by -6.625 points and NLL by +0.3197. At epoch five it reaches 78.649% / 0.9560, changing accuracy by -6.232 points and NLL by +0.3664 at identical exposure.

The head is one affine map from five concatenated 768-value pre-classifier vectors to 200 logits. It starts as the exact union of the frozen local classifiers and can then learn cross-node weights. Its 768,200 parameters are 93.6% fewer than the macro head's 12,055,496. Both conditions start from the same five source LoRAs and use the same data, augmentation, optimizer schedule, and checkpoint rule.

What the capacity controls establish

The rank-80 joint-IID control reaches **80.125% accuracy / 0.8339 NLL** at the fixed epoch-five endpoint: +2.263 accuracy points and -0.1085 NLL versus rank 16. The frontier remains 4.756 points higher and 0.2443 NLL lower.

Rank 80 and the frontier each expose 6,635,520 LoRA parameters and 60,970 image presentations. The frontier additionally trains a 12,055,496-parameter macro head, starts from five pretrained adapters, and executes five ViT paths; rank 80 starts one zero-effect adapter and uses one path. Its diagnostic minimum NLL is 0.7954 with 80.026% accuracy at epoch 3.

The rank-224 control reaches **81.109% accuracy / 0.7772 NLL** at epoch five. Its 18,674,812 trainable parameters are 16,204 (0.087%) fewer than the adaptive frontier's 18,691,016, which is the closest possible integer-rank match. Relative to rank 80, it changes accuracy by +0.984 percentage points and NLL by -0.0568. At matched exposure, the adaptive frontier remains 3.772 accuracy points higher, while the adaptive frontier is 0.1875 NLL lower. Its minimum NLL is 0.7199 at epoch 3; accuracy peaks at 82.060% at epoch 4. Validation NLL then rises by 0.0573 through epoch five while the training objective keeps improving, a pattern consistent with late overfitting or miscalibration under the inherited rank-16 schedule.

This removes active parameter count as a large numerical mismatch, but it does not match factorization, initialization, training history, or inference compute. Rank 224 is one fresh adapter in one ViT path; the frontier starts from five pretrained adapters and combines five ViT paths with a macro transformer.

The head schedule is the previous minimum-NLL winner: effective batch 64, peak AdamW LR 3e-5, 50 epochs, 5% warmup, and cosine decay. The LoRA peak LR 5e-4 is imported from the same-split joint recipe but run here in AdamW under the shared schedule. It has not been tuned for this coupled model.

Limits. This is a one-seed validation screen. The H cells change both unique data and total optimizer updates. Maximum accuracy is exploratory because all 50 validation checkpoints are visible. A promising condition needs replication and a focused LoRA/head learning-rate audit before any locked-test use.

Integrity and reuse

The training seals record zero fit-validation overlap, zero test overlap, zero test evaluations, and unchanged source hierarchy files. Fresh-process replay authenticated all six original macro-frontier cells, all eight architecture-sweep cells, and the full-fit controls without taking an optimizer step. Large checkpoints and model weights remain local; compact histories, tables, plots, and protocol records are the report surface.